Endpoint singularities and nonreal zeros in fractional theta deformations

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Abstract

We study the cosine transforms of the arithmetic theta kernel multiplied by $\exp(tu^{2a})$ on the positive half-line. For $1 < a < 2$ and $t \neq 0$, the transform is entire but has only finitely many real zeros and infinitely many nonreal zeros. The mechanism is the endpoint asymptotic $F_{a,t}(x) \sim tK(0)\Gamma(2a+1)\sin(\pi(a-1))x^{-2a-1}$. Near quadratic order a uniform absolute error gives a real-zero radius bounded by a constant times the inverse squared order gap, while full rectangular relative control fails. We give the canceled-derivative proof on both sides of the cubic endpoint and treat that endpoint separately. The transforms converge locally uniformly in the complex plane and uniformly on the entire real axis as the order varies. Nevertheless their global zero sets need not be real. At a limiting order with only real simple zeros, every selected nonreal-zero sequence escapes compact sets. The exact domain and quantifiers agree with a frozen Lean development. The Fourier endpoint mechanism has classical antecedents; novelty of the arithmetic specialization and its combination with order limits is not asserted.

1 The family and the main obstruction

For $u \geq 0$ set

$$K(u) = \sum_{n=1}^{\infty} \left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right) e^{-\pi n^2 e^{4u}}. \tag{1}$$

The kernel has an even analytic continuation near the real axis, whose restriction to negative real arguments we also denote by K. For $a > 0$, $t \in \mathbb{R}$, and $z \in \mathbb{C}$, define

$$F_{a,t}(z) = \int_0^{\infty} e^{tu^{2a}} K(u)\cos(zu)\,du. \tag{2}$$

The time sign is positive. Rapid kernel decay makes the integral meaningful for every real time. At integer $a = k$, the tilt is $\exp(tu^{2k})$; thus $a = 2$ is the quartic order, $a = 3$ is the sixth order, and $a = 3/2$ is the cubic endpoint between the ranges considered below. These parameter conventions will not be interchanged.

Theorem 1.1. *Fix $1 < a < 2$ and $t \neq 0$. The function $F_{a,t}$ is a nonzero even entire function, real on the real axis. As $x \to +\infty$,*

$$x^{2a+1}F_{a,t}(x) \longrightarrow L(a,t) := tK(0)\Gamma(2a+1)\sin(\pi(a-1)) \neq 0. \tag{3}$$

Its real zero set is finite, and its nonreal zero set is infinite. Consequently there is no real T such that all zeros of $F_{a,t}$ are real for every $t \geq T$.

Theorem 1.2. *For fixed t and $a_0 > 0$, $F_{a,t} \to F_{a_0,t}$ locally uniformly on $\mathbb{C}$ and uniformly on $\mathbb{R}$ as $a \to a_0$. If all zeros of $F_{a_0,t}$ are real and simple, and*

$$a_n \to a_0, \qquad F_{a_n,t}(z_n) = 0, \qquad \operatorname{Im} z_n \neq 0,$$

then $|z_n| \to \infty$. For each fixed $t \neq 0$, there is an escaping selection of nonreal zeros along $a_n = 1 + 1/(n+3)$ without any assumption about the zeros of $F_{1,t}$.

The two escape statements have different quantifiers. The last one chooses a sufficiently distant zero separately at each order. The preceding statement applies to *every* selection, and requires both reality and simplicity of the limiting zeros. No reality or simplicity hypothesis at quadratic order is silently deduced from convergence.

Pólya's work on entire Fourier transforms is the natural historical context [1]. The non-even-power examples identified in the literature already warn that endpoint regularity can affect global zero reality. Cardon's real-zero-preserving transform theorems [3] provide another comparison: their hypotheses are not replaced by positivity of the kernel in (2). The results here concern this precise arithmetic deformation and its formal verification. A bounded preliminary search has not settled priority for that specialization. Historical context is not used as a premise of the analytic proofs below.

2 Kernel identities and bounds

We first record why the kernel has the properties needed at both ends of the integral. They serve different purposes: evenness cancels odd derivatives at zero, whereas rapid decay controls every tilt and differentiation at infinity.

Lemma 2.1. *The kernel is positive on $[0,\infty)$, is even and real analytic on the real line, and for every integer $j \geq 0$ satisfies a bound*

$$|K^{(j)}(u)| \leq C_j \exp\left(C_j u - \frac{\pi}{2} e^{4u}\right) \qquad (u \geq 0) \tag{4}$$

for a finite positive C_j. In particular $K(0) > 0$ and every odd derivative of K vanishes at zero.

Proof. Every summand in (1) is positive on $u \geq 0$: factoring out $\pi n^2 e^{5u}$ leaves $2\pi n^2 e^{4u} - 3 > 0$. The defining series converges absolutely. After any fixed number of derivatives, each summand is a finite sum of terms bounded by

$$C n^M e^{Mu} e^{-\pi n^2 e^{4u}}$$

for some integer M depending on the derivative order. With $s = e^{4u} \geq 1$, the inequality $n^2 s \geq (n^2 + s)/2$ shows that their sum is at most

$$C e^{Mu - \pi s/2} \sum_{n \geq 1} n^M e^{-\pi n^2/2}.$$

The numerical series is finite. This proves (4) after enlarging the constant and justifies termwise differentiation on compact intervals.

For evenness, let $\vartheta(v) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 v}$ and $Q(u) = e^u \vartheta(e^{4u})$. The theta transformation $\vartheta(v) = v^{-1/2}\vartheta(1/v)$ gives $Q(-u) = Q(u)$. Differentiating the absolutely convergent series shows

$$K(u) = \frac{1}{16}(Q''(u) - Q(u)). \tag{5}$$

Hence K is even. The theta transformation itself follows from Poisson summation of the Gaussian: periodize $e^{-\pi v x^2}$, whose nth Fourier coefficient is $v^{-1/2}e^{-\pi n^2/v}$, and evaluate the absolutely convergent Fourier series at zero. Absolute and locally uniform convergence justify this computation. The same defining theta series is holomorphic where $\mathrm{Re}(e^{4u}) > 0$, in particular on the strip $|\operatorname{Im} u| < \pi/8$. It follows that K is analytic near the real line. Evenness gives the odd-derivative cancellations. □

The Gaussian Fourier coefficient in this proof follows by scaling the real Gaussian integral and differentiating its Fourier transform: integration by parts gives $\widehat{g}'(\xi) = -2\pi\xi\widehat{g}(\xi)/v$, and $\widehat{g}(0) = v^{-1/2}$. This determines the coefficient and fixes the normalization in (5).

Lemma 2.2. *For every $a > 0$ and $t \in \mathbb{R}$, the integral (2) is entire, even, obeys $F_{a,t}(\bar{z}) = \overline{F_{a,t}(z)}$, and has $F_{a,t}(0) > 0$. Moreover, for some $C = C(a,t) > 0$,*

$$|F_{a,t}(z)| \le C\exp(|z|^{3/2}) \qquad (z \in \mathbb{C}). \tag{6}$$

Proof. For $|z| \le R$, the jth spatial derivative of the cosine is bounded by $u^j e^{Ru}$. Bound (4) makes $u^j \exp(tu^{2a} + Ru)K(u)$ absolutely integrable for every j. Differentiation under the integral therefore gives entireness. Evenness and conjugation symmetry follow from cosine, and positivity at zero follows from the positive density.

For the growth bound put $r = |z|$. The elementary inequality $ru \le r^{3/2} + u^3$ holds for $r, u \ge 0$: if $u \le \sqrt{r}$ use the first term, and otherwise use the second. Thus

$$|F_{a,t}(z)| \le e^{r^{3/2}} \int_0^\infty \exp(tu^{2a} + u^3)K(u)\,du.$$

The integral is a finite positive constant by the same decay bound. □

The exponent $3/2$ is a convenient subquadratic bound, not an assertion of the exact entire order. Its strict inequality below two will be enough for the finite-zero rigidity argument.

3 The endpoint Fourier calculation

Write $\beta = 2a \in (2,4)$ and $g(u) = e^{tu^\beta}K(u)$ on $u > 0$. All derivatives used below are ordinary derivatives of this actual positive-axis function. We do not differentiate a truncated formal jet beyond the orders for which it represents those derivatives.

For $b > 0$, the Laplace integral gives

$$\int_0^\infty u^{b-1}e^{-u}e^{ixu}\,du = \Gamma(b)(1-ix)^{-b}. \tag{7}$$

The right side uses the branch with $\arg(1-ix) \in (-\pi/2, \pi/2)$. Indeed, for positive real w the identity $\int u^{b-1}e^{-wu}\,du = \Gamma(b)w^{-b}$ follows by substitution. Both sides are holomorphic for $\operatorname{Re} w > 0$, by dominated differentiation on compact subsets, so the identity theorem extends it to that half-plane and hence to $w = 1 - ix$. For $x \to +\infty$, taking imaginary and real parts yields

$$x^b \int_0^\infty u^{b-1}e^{-u}\sin(xu)\,du \longrightarrow \Gamma(b)\sin(\pi b/2), \tag{8}$$

$$x^b \int_0^\infty u^{b-1}e^{-u}\cos(xu)\,du \longrightarrow \Gamma(b)\cos(\pi b/2). \tag{9}$$

These formulas replace a heuristic oscillatory integral of a bare power by an absolutely convergent integral with a known transform. The subtraction of the exponential cutoff is what makes the actual remainder differentiable with an integrable derivative.

3.1 Cancellation estimates near zero

Use the falling factorial $(\beta)_m = \beta(\beta-1)\cdots(\beta-m+1)$. For $m = 3$ or 4, put

$$C_m = tK(0)(\beta)_m, \qquad R_m(u) = g^{(m)}(u) - C_m u^{\beta-m} e^{-u}. \tag{10}$$

The following estimates are for fixed β, t; no uniformity in the order parameter is being asserted.

Lemma 3.1. *If $2 < \beta < 3$, then $R_3, R_3' \in L^1(0,\infty)$, $R_3(0+) = 0$, and $R_3(u) \to 0$ as $u \to \infty$. If $3 < \beta < 4$, the same integrability and decay statements hold for R_4, R_4', with $R_4(0+) = K^{(4)}(0)$ in place of zero.*

Proof. Near zero, expand only the regular factors:

$$K(u) = K(0) + \frac{K''(0)}{2}u^2 + O(u^4), \qquad e^{tu^\beta} = 1 + tu^\beta + O(u^{2\beta}).$$

The estimates persist after any of the finitely many positive-axis derivatives required here. One justification is to write the exponential remainder as $u^{2\beta}h(u^\beta)$ with h analytic near zero, and the kernel remainders as Taylor remainders with smooth factors. Repeated product differentiation then gives, for $m = 3, 4$,

$$g^{(m)}(u) = K^{(m)}(u) + C_m u^{\beta-m} + O(u^{\beta+2-m}) + O(u^{2\beta-m}). \tag{11}$$

Its next derivative has the differentiated power bounds, by that same factorization argument, rather than by differentiating an unqualified big-O statement.

For $m = 3$ the singular part left after the cutoff is $C_3 u^{\beta-3}(1 - e^{-u}) = O(u^{\beta-2})$. The kernel term $K'''(u)$ is $O(u)$ by evenness. Thus $R_3(0+) = 0$, and locally

$$R_3'(u) = O(1) + O(u^{\beta-3}) + O(u^{2\beta-4}).$$

All exponents exceed -1 when $\beta > 2$.

For $m = 4$ the leftover cutoff term is $O(u^{\beta-3})$, and the kernel term tends to $K^{(4)}(0)$. Its derivative satisfies

$$R_4'(u) = O(1) + O(u^{\beta-4}) + O(u^{2\beta-5}),$$

whose exponents exceed -1 when $\beta > 3$. The remainders themselves are locally integrable as well. At infinity, repeated product differentiation gives finite sums of powers of u times $e^{tu^\beta}K^{(j)}(u)$; all are integrable and tend to zero by (4). The cutoff and its derivative have the same properties at infinity. □

In particular the fourth-order remainder need not vanish at zero. The forthcoming cosine integration by parts uses a sine boundary factor, which kills its finite endpoint value. Assuming a zero endpoint for that remainder would be an unnecessary and unjustified strengthening.

3.2 The range $2 < \beta < 3$

Since $g'(0+) = 0$, three integrations by parts give

$$x^3 F_{a,t}(x) = \int_0^\infty g'''(u)\sin(xu)\,du \qquad (x > 0). \tag{12}$$

For completeness, the first step is $F = -x^{-1}\int g'\sin(xu)$; the second produces $\int g'\sin(xu) = x^{-1}\int g''\cos(xu)$ because $g'(0+) = 0$; the third gives $\int g''\cos(xu) = -x^{-1}\int g'''\sin(xu)$. All derivatives through the third order are integrable. Their positive-axis endpoint limits and rapid decay justify the boundary terms by first integrating over $[\varepsilon, M]$ and passing to the limits.

By Lemma 3.1, another integration by parts gives

$$x\int_0^\infty R_3(u)\sin(xu)\,du = \int_0^\infty R_3'(u)\cos(xu)\,du.$$

The right side is bounded by $\|R_3'\|_{L^1}$; here and below $\|\cdot\|_{L^1}$ denotes the integral of the absolute value. Set $b = \beta - 2 \in (0,1)$. Combining (12) and (10), multiplying by x^b, and using (8), we obtain

$$x^{\beta+1}F_{a,t}(x) \longrightarrow C_3\Gamma(\beta-2)\sin(\pi(\beta-2)/2).$$

The remainder vanishes because $x^bO(x^{-1}) \to 0$. The recurrence for Gamma gives $(\beta)_3\Gamma(\beta-2) = \Gamma(\beta+1)$, and $\sin(\pi(\beta-2)/2) = \sin(\pi(a-1))$. This proves (3) in the first range.

3.3 The range $3 < \beta < 4$

Here $g'(0+) = g'''(0+) = 0$. The previous integrations and one more give

$$x^4F_{a,t}(x) = \int_0^\infty g^{(4)}(u)\cos(xu)\,du. \tag{13}$$

Lemma 3.1 now yields

$$x\int_0^\infty R_4(u)\cos(xu)\,du = -\int_0^\infty R_4'(u)\sin(xu)\,du.$$

At zero the boundary term is $R_4(0+)\sin 0 = 0$; no vanishing of $R_4(0+)$ is required. Set $b = \beta - 3 \in (0,1)$ and use (9). The limit in (13) becomes

$$C_4\Gamma(\beta-3)\cos(\pi(\beta-3)/2) = tK(0)\Gamma(\beta+1)\sin(\pi(a-1)).$$

Again the remainder tends to zero after scaling because $b < 1$. This proves (3) in the second range.

3.4 The cubic endpoint $\beta = 3$

The positive-axis density is now $g(u) = e^{tu^3}K(u)$, smooth up to zero on that half-line. Evenness of K gives $g'(0) = 0$ and $g'''(0) = 6tK(0)$. The four-step integration by parts retains that endpoint term:

$$x^4F_{3/2,t}(x) = 6tK(0) + \int_0^\infty g^{(4)}(u)\cos(xu)\,du. \tag{14}$$

Since $g^{(4)} \in L^1$, the last integral tends to zero by the Riemann–Lebesgue lemma. This agrees with (3) because $\Gamma(4) = 6$ and $\sin(\pi/2) = 1$.

One cannot obtain this case by substituting $b = 0$ into the preceding Gamma-cutoff argument: the cutoff $u^{b-1}e^{-u}$ would cease to be integrable at zero. The separate endpoint calculation is essential.

4 From the asymptotic to the global zero obstruction

The coefficient in (3) is nonzero: $K(0) > 0$, $\Gamma(2a+1) > 0$, and $\sin(\pi(a-1)) > 0$ for $1 < a < 2$. Thus the asymptotic excludes zeros for all sufficiently large positive real x. Evenness excludes them for sufficiently negative x. On the remaining compact real interval, a nonzero entire function has finitely many zeros. This proves finiteness of the real zero set.

To deduce nonreal infinitude we must also prove that the whole zero set is infinite. Nonpolynomiality alone is insufficient: exponential functions are nonpolynomial and zero free. The subquadratic growth bound supplies the necessary additional rigidity.

Lemma 4.1 (Finite-zero rigidity)**.** *Let f be a nonzero entire function satisfying $|f(z)| \le Ce^{|z|^{3/2}}$ for some $C > 0$. If f has finitely many zeros, then $f(z) = P(z)e^{\alpha z}$ for a nonzero complex polynomial P and $\alpha \in \mathbb{C}$. Such a function cannot tend to zero along both real tails.*

Proof. Remove every zero with its multiplicity, forming a finite polynomial P_0. Removable singularities make $h = f/P_0$ a zero-free entire function. A nonzero polynomial has a positive lower bound outside a sufficiently large disk. On the complementary disk h is bounded by continuity. Therefore $|h(z)| \le De^{|z|^{3/2}}$ for some $D > 0$. This justification for the quotient bound is needed; it does not follow from a numerator bound without controlling the denominator.

The zero-free entire function h has an entire logarithm ℓ. For instance, integrate the entire function h'/h on the simply connected plane and choose the additive constant at zero. Then $\operatorname{Re}\ell(z) \le \log D + |z|^{3/2}$. The Borel–Carathéodory inequality on disks of radii R and $2R$ gives

$$\max_{|z|\le R} |\ell(z) - \ell(0)| = O(1 + R^{3/2}).$$

Cauchy's estimates show for every integer $m \ge 2$ that $|\ell^{(m)}(0)| \le m!O(1+R^{3/2})/R^m \to 0$. Thus the Taylor series is affine, $\ell(z) = \ell(0) + \alpha z$. Absorb $e^{\ell(0)}$ into P_0 to obtain the claimed factorization.

Choose a sign $s \in \{1, -1\}$ with $s \operatorname{Re}\alpha \ge 0$. Along $z = sx$, $x \to +\infty$, the exponential has modulus at least one and the nonzero polynomial has a positive lower bound. Hence $f(sx)$ cannot tend to zero. □

The Borel–Carathéodory estimate used here is the disk inequality

$$\max_{|z|\le r} |h(z) - h(0)| \le \frac{2r}{R-r} \left(\max_{|z|\le R} \operatorname{Re} h(z) - \operatorname{Re} h(0) \right), \quad 0 < r < R,$$

for holomorphic h near the closed disk. It follows by applying the maximum principle and Schwarz's lemma to $(h - h(0))/(2M - (h - h(0)))$, where $M > \max \operatorname{Re}(h - h(0))$, and then letting M decrease. This also explains why a bound on the real part suffices for the Taylor-coefficient argument.

Completion of Theorem 1.1. The density in (2) is integrable, so its cosine transform tends to zero as $x \to \pm\infty$. This form of the Riemann–Lebesgue lemma follows by approximating the density in L^1 by finite step functions with bounded interval support: their cosine integrals decay, and the error is bounded by the L^1 distance uniformly in x. Lemma 2.2 and Lemma 4.1 now imply that the total complex zero set is infinite. Removing the finite real zero set leaves infinitely many nonreal zeros.

If an eventual all-real starting time T existed, choose $t = \max\{T, 1\} > 0$. The preceding conclusion at this nonzero time contradicts the required reality of every zero. □

Corollary 4.2. *For $1 < a < 2$ and $t \neq 0$, $F_{a,t}$ is not a locally uniform limit on $\mathbb{C}$ of real polynomials all of whose roots are real.*

Proof. On the upper half-plane such polynomials are zero free, except that an identically zero polynomial may occur. Local convergence to the nonzero function $F_{a,t}$ excludes identically zero polynomials eventually. Hurwitz's theorem then makes the limit zero free there unless it is identically zero. The same argument applies to the lower half-plane. Either conclusion contradicts a nonreal zero of $F_{a,t}$. □

Only the necessary direction of the polynomial-limit criterion is used. No general converse factorization theorem for the Laguerre–Pólya class is needed for this obstruction.

5 Order convergence on compact sets and on the real axis

Fix t and $a_0 > 0$. In a sufficiently small interval around a_0, choose $0 < A \leq a \leq B$. For $u \geq 1$, $u^{2a} \leq u^{2B}$, and for $0 \leq u \leq 1$ it is bounded by one. For $|z| \leq R$, every fixed spatial derivative of the integrand is bounded by an integrable multiple of

$$(1 + u^j) \exp\big(\max(t, 0)(1 + u^{2B}) + Ru\big)K(u).$$

Continuity of $a \mapsto u^{2a}$ for $u > 0$, together with dominated convergence, therefore gives

$$\int_0^\infty u^j e^{Ru} |e^{tu^{2a}} - e^{tu^{2a_0}}| K(u)\, du \longrightarrow 0. \tag{15}$$

At $u = 0$ the integrand is defined by continuity; its value on that single point does not affect the integral.

Taking $j = 0$ bounds the supremum of $|F_{a,t} - F_{a_0,t}|$ on $|z| \leq R$, proving locally uniform convergence. Taking $j = 1$ proves local uniform convergence of the spatial derivatives. Thus values and derivatives are jointly continuous in order and spatial argument near any positive order.

For real x, $|\cos(xu)| \leq 1$, giving the stronger domain bound

$$\sup_{x \in \mathbb{R}} |F_{a,t}(x) - F_{a_0,t}(x)| \leq \int_0^\infty |e^{tu^{2a}} - e^{tu^{2a_0}}| K(u)\, du \longrightarrow 0. \tag{16}$$

This is uniform convergence on the entire real axis at fixed time. It is not uniform convergence on the entire complex plane. The factor $e^{|\operatorname{Im} z| u}$ would prevent that inference from (16).

These convergence results coexist with Theorem 1.1: every nearby fractional order can have infinitely many nonreal zeros, even when a particular limiting transform has none. Neither small errors on the real axis nor convergence on each fixed disk controls the complete complex zero set at one common parameter neighborhood.

6 Simple real limits force compact escape

We give the local mechanism explicitly, so the simplicity hypothesis does not disappear behind a qualitative reference to continuity of roots.

Lemma 6.1. *Suppose $a_n \to a_0 > 0$, $z_n \to w \in \mathbb{R}$, $F_{a_n,t}(z_n) = 0$, and $\operatorname{Im} z_n \neq 0$ for every n. Then $F_{a_0,t}(w) = 0$ and $F'_{a_0,t}(w) = 0$.*

Proof. Joint continuity gives the value assertion. Suppose the derivative at w were $c \neq 0$. Joint continuity of the derivative gives a disk D about w and an order neighborhood such that

$$|F'_{a,t}(z) - c| \leq |c|/2 \qquad (z \in D)$$

for all those orders. Since D is convex, integration along the segment between $z, v \in D$ shows

$$|F_{a,t}(z) - F_{a,t}(v) - c(z - v)| \leq \frac{|c|}{2}|z - v|.$$

Hence $|F_{a,t}(z) - F_{a,t}(v)| \geq |c||z - v|/2$. For all large n, both z_n and $\bar{z}_n$ lie in D. Conjugation symmetry makes both zeros of $F_{a_n,t}$. The lower bound forces $z_n = \bar{z}_n$, a contradiction. □

Proposition 6.2. *If every zero of $F_{a_0,t}$ is real and simple, then for each compact set $Q \subset \mathbb{C}$ there is an order neighborhood of a_0 in which every zero belonging to Q is real.*

Proof. Otherwise select $a_n \to a_0$ and nonreal zeros $z_n \in Q$. Compactness supplies a subsequence converging to $w \in Q$. Joint continuity makes w a limiting zero; the hypothesis makes it real. Lemma 6.1 makes it multiple, contrary to simplicity. □

Proof of the escape assertions in Theorem 1.2. Apply Proposition 6.2 to each closed disk $\{|z| \leq R\}$. A sequence of nonreal zeros with $a_n \to a_0$ eventually avoids that disk. This is exactly $|z_n| \to \infty$.

For the unconditional selection near one, put $a_n = 1 + 1/(n+3) \in (1, 3/2)$. Theorem 1.1 gives an infinite nonreal zero set for each a_n. The nonzero entire function has only finitely many zeros in any compact disk, so choose one with $|z_n| > n$. This chosen sequence escapes and needs no hypothesis about $F_{1,t}$. □

The simplicity assumption cannot be dropped in the general compact stability principle. For instance, $z^2 + \varepsilon$ converges locally uniformly to z^2 as $\varepsilon \downarrow 0$, while its nonreal zeros converge to the double real zero at zero. This elementary comparison is explanatory; it is not a claim that this polynomial family is an arithmetic theta flow.

7 Endpoints, density analyticity, and parameter limits

At $t = 0$, every order in (2) gives the same transform $\int K(u)\cos(zu)\,du$. The coefficient $L(a, 0)$ vanishes, so the preceding nonreal-zero obstruction makes no assertion at that time. In particular no conclusion about the Riemann hypothesis is extracted from a theorem whose hypothesis is $t \neq 0$.

Likewise $L(1, t) = L(2, t) = 0$. Direct differentiation of the explicit coefficient gives

$$\frac{L(a, t)}{a - 1} \longrightarrow 2\pi t K(0) \qquad (a \downarrow 1). \tag{17}$$

Indeed, the sine factor has derivative π at one and $\Gamma(2 \cdot 1 + 1) = 2$; derivatives of the other factors multiply the vanishing sine. This is a coefficient limit at fixed time, not a uniform relative asymptotic on the rectangle $a \downarrow 1$, $x \to \infty$.

The density distinction is also exact. At $a = 3/2$ its even full-line form is $e^{t|u|^3}K(u)$. For $t \neq 0$ the right and left third derivatives at zero differ by $12tK(0)$, so this full-line density cannot be the restriction of a holomorphic germ through zero. Its positive-axis form $e^{tu^3}K(u)$ does have a

holomorphic extension there. These are different functions on the negative axis. Nevertheless its Fourier transform is entire by Lemma 2.2. Entire dependence on frequency follows from decay and does not require analyticity of the full-line density at its endpoint.

The same distinction prevents an unjustified complex branch choice for a nonintegral real exponent. Definition (2) uses the real power of a nonnegative real variable. The proof of its entireness differentiates in z, not in a fictitious complex continuation of u^{2a} across zero.

The library separately proves eventual real and simple zeros at quartic order. Combined with the compact escape principle, that result gives an application with $a_n = 2 - 1/(n+3)$. Its global quartic saddle analysis is a separate theorem, not reconstructed in this paper and not an assumption in Theorem 1.1. No effective upper time, nonreal-zero escape rate, or classification for real orders above two follows from the present argument.

8 Uniform absolute errors and a real-frequency transition

The fixed-order cancellation estimates of Section 3 do not alone give uniformity as $a \downarrow 1$. The coefficient of their singular derivative must be retained. This leads to a quantitative real-frequency statement, while still excluding relative uniformity on a full rectangle.

Theorem 8.1. *For each fixed real t, there is $A_t \geq 0$ such that*

$$|x^{2a+1}F_{a,t}(x) - L(a,t)| \leq A_t x^{-1/2} \quad \left(1 < a \leq \frac{5}{4},\ x \geq 1\right). \tag{18}$$

If $t \neq 0$, there are $0 < \delta_t \leq 1/4$, $C_t \geq 0$ and $K_t > 0$ such that, for $1 < a < 1 + \delta_t$,

$$\left|\frac{x^{2a+1}F_{a,t}(x)}{L(a,t)} - 1\right| \leq \frac{C_t}{(a-1)\sqrt{x}} \quad (x \geq 1), \tag{19}$$

$$F_{a,t}(x) = 0,\ x \in \mathbb{R} \quad \Longrightarrow \quad |x| \leq \max\left\{1, \left(\frac{K_t}{a-1}\right)^2\right\}. \tag{20}$$

In particular the relative asymptotic holds along every sequence $a_n > 1$, $a_n \to 1$, $x_n \to \infty$ with $(a_n - 1)\sqrt{x_n} \to \infty$.

A surviving boundary mass behind pointwise vanishing

There is an exact measure-level reason to keep the small coefficient in the cutoff. For $a > 1$, put $b = 2a - 2 > 0$ and define the real signed density on $u > 0$

$$c_{a,t}(u) = tK(0)(2a)(2a-1)(2a-2)u^{2a-3}e^{-u}, \qquad m(a,t) = tK(0)\Gamma(2a+1).$$

This is precisely the real value of the cutoff subtracted from the third derivative, not the full deformed arithmetic density. The Gamma integral and its recurrence give

$$\int_0^\infty c_{a,t}(u)\,du = m(a,t), \qquad \int_0^\infty |c_{a,t}(u)|\,du = |m(a,t)|, \tag{21}$$
$$\int_0^\infty u|c_{a,t}(u)|\,du = b|m(a,t)|.$$

Indeed the integrals of $u^{b-1}e^{-u}$ and $u^b e^{-u}$ are $\Gamma(b)$ and $\Gamma(b+1) = b\Gamma(b)$, respectively; three recurrences yield $b(b+1)(b+2)\Gamma(b) = \Gamma(b+3) = \Gamma(2a+1)$. All coefficients other than t are

positive. This proves both the signed and absolute formulas, as well as integrability and the first absolute moment.

`RiemannRealOrderCutoffBoundary.cutoff_integral`.

`RiemannRealOrderCutoffBoundary.cutoff_norm_integral`.

`RiemannRealOrderCutoffBoundary.cutoff_norm_first_moment`.

If $f : \mathbb{R} \to \mathbb{C}$ is globally Lipschitz with constant L, then $|f(u)| \le |f(0)| + Lu$ on the positive half-line, so the preceding moments justify every integral in

$$\left| \int_0^\infty c_{a,t}(u) f(u)\, du - m(a,t) f(0) \right| \le Lb|m(a,t)|. \tag{22}$$

To prove it, subtract $f(0)$ under the integral, use $|f(u) - f(0)| \le Lu$, and apply the last identity in (21). Since Gamma is continuous at 3, $m(a,t) \to 2tK(0)$ as $a \downarrow 1$; hence

$$\int_0^\infty c_{a,t}(u) f(u)\, du \longrightarrow 2tK(0) f(0). \tag{23}$$

This holds also for unbounded Lipschitz tests, and for either sign of t. At $t = 0$ the density and all the displayed quantities vanish.

`RiemannRealOrderCutoffConcentration.cutoff_lipschitz_error_bound`.

`RiemannRealOrderCutoffConcentration.cutoff_lipschitz_tendsto`.

For $t \neq 0$ the normalized density is exactly the positive Gamma probability density

$$\frac{c_{a,t}(u)}{m(a,t)} = \frac{u^{b-1} e^{-u}}{\Gamma(b)}.$$

Let ν_a denote this probability on $[0, \infty)$, with no atom at zero. Its first moment is b. The preceding calculation therefore gives the exact human consequence

$$\sup_{\mathrm{Lip}(f) \le 1} \left| \int f\, d\nu_a - f(0) \right| = b, \tag{24}$$

where tests are real functions on $[0, \infty)$ with no boundedness restriction. The upper bound is the first-moment inequality; the test $f(u) = u$ attains it. Integrals exist because such tests grow at most linearly. This is an explicit Gamma approximate identity, not a new probability law associated with the entire transform.

It also converges against every bounded continuous test, a human consequence beyond the named globally Lipschitz formulation. For $\eta > 0$, the first moment implies $\nu_a([\eta, \infty)) \le b/\eta$. Splitting the test error at η gives

$$\left| \int f\, d\nu_a - f(0) \right| \le \sup_{0 \le u \le \eta} |f(u) - f(0)| + 2\|f\|_\infty b/\eta.$$

First let $a \downarrow 1$ at fixed η, then let $\eta \downarrow 0$. Thus the signed cutoff measures converge weakly on the *closed* half-line to $2tK(0)\delta_0$. The endpoint is included in this statement; the original density integrals themselves remain over $u > 0$.

Nevertheless $c_{a,t}(u) \to 0$ for every fixed $u > 0$: the factor $2a - 2$ tends to zero and the other factors have finite limits there. For $t \neq 0$, (21) instead gives $\int |c_{a,t}| \to |2tK(0)| > 0$, because $K(0) > 0$. There is no common integrable dominator on any order neighborhood $1 < a < 1 + \delta$. Otherwise take the sequence $a_n = 1 + \delta/(n+2)$; the countable intersection of its almost-everywhere domination events and dominated convergence would force its integrals to zero, contradicting the

mass limit. This is precisely why pointwise disappearance cannot remove this cutoff from the parameter-limit analysis. It does not contradict the already proved uniform convergence of the original transforms, nor does it alter any zero classification.
`RiemannRealOrderCutoffConcentration.cutoff_pointwise_tendsto_zero`.
`RiemannRealOrderCutoffConcentration.no_common_integrable_majorant`.

A uniform integrable derivative after cancellation

Fix t, put $\beta = 2a$, $b = \beta - 2 \in (0, 1/2]$ and $E(u) = e^{tu^\beta}$. Keep R_3 and C_3 from (10). Write

$$p_j(u) = t(\beta)_j u^{\beta-j}, \qquad D(u) = E(u)K(u) - K(0)e^{-u}.$$

All constants in this subsection may depend on t, K, but are chosen before a. On $0 < u \le 1$, E, p_1, p_2 are uniformly bounded. The derivative bound on K and the mean-value estimates $|e^v - 1| \le |v|e^{|v|}$ and $1 - e^{-u} \le u$ give $|D(u)| \le Cu$: use

$$D = E(K - K(0)) + (E - 1)K(0) + K(0)(1 - e^{-u}), \quad u^\beta \le u.$$

Since $2 < \beta \le 5/2$,

$$|p_3(u)| \le Cbu^{b-1}, \qquad |p_4(u)| \le Cbu^{b-2}, \qquad \int_0^1 bu^{b-1}\,du = 1. \tag{25}$$

The last equality follows by evaluating u^b at the endpoints; the small factor b must not be discarded.

For clarity the entire fourth-derivative polynomial is

$$\begin{aligned} g^{(4)} = E\big[&K^{(4)} + 4p_1K^{(3)} + 6(p_2 + p_1^2)K'' + 4(p_3 + 3p_1p_2 + p_1^3)K' \\ &+ (p_4 + 4p_1p_3 + 3p_2^2 + 6p_1^2p_2 + p_1^4)K\big]. \end{aligned}$$

It follows by repeatedly differentiating $g = EK$, using $E' = p_1E$ and $p_j' = p_{j+1}$. Remove just the term Ep_4K and call the remaining expression $H(u)$. Every remaining occurrence of p_3 is linear with uniformly bounded coefficient. All other factors are uniformly bounded on $(0, 1]$, so $|H(u)| \le C(1 + |p_3(u)|)$. Differentiating the actual cutoff $C_3u^{\beta-3}e^{-u} = p_3K(0)e^{-u}$ gives the exact cancellation

$$R_3'(u) = p_4(u)D(u) + H(u) + p_3(u)K(0)e^{-u}.$$

Thus (25) bounds its norm by $C(1 + bu^{b-1})$ on $(0, 1]$, whose integral is bounded uniformly in b. No common pointwise integrable majorant for bu^{b-1} is asserted or needed.

On $u \ge 1$, the finitely many phase derivatives have a common polynomial bound in u, and $E(u) \le e^{|t|u^{5/2}}$. Lemma 2.1 bounds every $K^{(j)}$, $j \le 4$, by $C_j \exp(C_ju - ce^{4u})$ for $c > 0$. Multiplication by the polynomial phase factors therefore leaves a common integrable bound, since e^{4u} dominates $u^{5/2}$ and every linear or logarithmic term. The cutoff derivative is bounded by Ce^{-u}: its powers are $u^{b-1}, u^{b-2} \le 1$ and its coefficients are uniformly bounded. Combining the two regions proves

$$\sup_{1<a\le 5/4} \int_0^\infty |R_3'(u)|\,du < \infty. \tag{26}$$

For each order the endpoint and infinity values of R_3 vanish, as proved in Lemma 3.1. One integration by parts therefore yields

$$\left| \int_0^\infty R_3(u) \sin(xu)\,du \right| \le \frac{M}{x}.$$

Multiplying by x^b gives at most $Mx^{b-1} \le Mx^{-1/2}$ for $x \ge 1$. This is the classical derivative-variation error mechanism, also recorded in DLMF [4, §2.3(i)]; the coefficient-aware endpoint estimate above is what makes its constant uniform here.

The cutoff transform has a uniform reciprocal-frequency rate

Let $m_t(\beta) = tK(0)\Gamma(\beta+1)$, continuous and bounded for $2 \le \beta \le 5/2$. Only this coefficient and the power profile are extended to $b = 0$ for bounding purposes; the actual cutoff integral below still has $b > 0$. Equation (7) and the Gamma recurrence give, for $x > 0$,

$$x^b C_3 \int_0^\infty u^{b-1} e^{-u} e^{ixu}\, du = m_t(\beta) B(1/x)^b, \qquad B(y) = (y-i)^{-1}.$$

The power is the principal one. On $0 \le y \le 1$, $B(y)$ stays in the open upper half-plane, $1/\sqrt{2} \le |B(y)| \le 1$ and $|B'(y)| \le 1$. For $0 \le b \le 1/2$, differentiation gives

$$\left| \frac{d}{dy} B(y)^b \right| = |bB(y)^{b-1} B'(y)| \le \frac{\sqrt{2}}{2}.$$

The branch is valid throughout the segment, so integration of this derivative gives $|B(1/x)^b - i^b| \le C/x$ uniformly in b for $x \ge 1$. Taking imaginary parts, the limit value is

$$m_t(\beta) \sin(\pi b/2) = L(a,t).$$

The cutoff error is therefore at most Q_t/x uniformly in order. The identity from Section 3 is

$$x^3 F_{a,t}(x) = \int_0^\infty g^{(3)}(u) \sin(xu)\, du.$$

Together with (26), it proves (18), since $x^{-1} \le x^{-1/2}$.

Division by the vanishing coefficient and its limitation

For $t \ne 0$, (17) gives constants $c_t > 0$ and $0 < \delta_t \le 1/4$ with $|L(a,t)| \ge c_t(a-1)$ for $1 < a < 1+\delta_t$. Divide (18) by this positive lower bound to obtain (19) with $C_t = A_t/c_t$. If $(a-1)\sqrt{x} > K_t := C_t + 1$, the right side is less than one, whereas a zero of the flow would make the left side equal one. Evenness supplies the same conclusion for negative real x. Squaring the resulting bound on $\sqrt{|x|}$, and treating $|x| < 1$ separately, proves (20). The diagonal relative limit follows directly from the bound. The constants exist for each fixed t; no numerical values, optimal power, complex-zero radius or uniformity in t is claimed.

The full rectangular relative assertion is actually false. For any $\delta > 0$, real R and finite M, there are $1 < a < 1+\delta$ with $a < 3/2$ and $x \ge \max(R, 1)$ such that

$$\left| \frac{x^{2a+1} F_{a,t}(x)}{L(a,t)} - 1 \right| > M \qquad (t \ne 0). \tag{27}$$

Indeed $F_{1,t}$ is a nonzero entire function: its value at zero is the positive integral of $e^{tu^2} K(u)$. Its real zeros are isolated, so choose such an x with $F_{1,t}(x) \ne 0$. At that fixed x, continuity in order gives $x^{2a+1} F_{a,t}(x) \to x^3 F_{1,t}(x) \ne 0$, while $L(a,t) \to 0$ and is nonzero for $1 < a < 3/2$. The absolute relative error therefore tends to infinity as $a \downarrow 1$, proving (27). This explains why a parameter-dependent region is needed; it does not assert that the particular transition scale in (19) is sharp.

9 Gaussian translation needs control beyond compact sets

The compact convergence proved above has a precise use in local zero arguments. It does not, by itself, justify passage through a Gaussian average along an unbounded imaginary line. We record a positive identity for the actual quadratic flow and a separate polynomial counterexample to that proposed continuity inference.

For $s > 0$ let $\nu_s = N(0, 2s)$, with density $(4\pi s)^{-1/2} e^{-y^2/(4s)}$; put $\nu_0 = \delta_0$. Completing the Gaussian square gives

$$\int_{\mathbb{R}} e^{by}\, d\nu_s(y) = e^{sb^2} \quad (b \in \mathbb{R}).$$

The complex version is $\int e^{qy}\, d\nu_s = e^{sq^2}$. For completeness, this follows for real q by the same square completion; both sides are entire in q. Differentiation under the integral is justified on $|q| \le R$ by the integrable bounds $|y|^j e^{R|y|}$ against the Gaussian. The identity theorem supplies the complex equality. At $s = 0$ it is immediate.

Proposition 9.1. *For every real t, $s \ge 0$ and $z \in \mathbb{C}$, the following are genuine absolutely integrable Gaussian averages:*

$$\int F_{1,t}(z + iy)\, d\nu_s(y) = F_{1,t+s}(z), \qquad \int F_{1,t}(z + y)\, d\nu_s(y) = F_{1,t-s}(z). \tag{28}$$

Proof. For the imaginary shift, the absolute integrand in the double cosine integral is bounded by

$$e^{tu^2} K(u) e^{|\operatorname{Im} z|u} e^{u|y|}.$$

Since $e^{u|y|} \le e^{uy} + e^{-uy}$, its integral in y is at most $2e^{(t+s)u^2 + |\operatorname{Im} z|u} K(u)$. This is integrable in $u \ge 0$ by Lemma 2.1; in particular product integrability is proved before Fubini is used. Expressing the cosine as the mean of two complex exponentials and applying the Gaussian formula multiplies it by e^{su^2}, giving the first identity. For the real shift, the absolute bound is already $e^{tu^2 + |\operatorname{Im} z|u} K(u)$, independent of y. The Gaussian characteristic formula multiplies the cosine by e^{-su^2}, giving the second identity. At $s = 0$ both reduce to evaluation. □

These are classical Gaussian translation identities with the actual arithmetic integrability checked. Their time signs differ because the displacement is imaginary or real. They give no conclusion about quadratic zero reality without a further real-zero theorem.

Define the time-one imaginary Gaussian functional on integrable entire functions by

$$\mathcal{H} f = \int_{\mathbb{R}} f(iy)\, d\nu_1(y).$$

For $m \ge 1$ put

$$q_m(z) = (-1)^m \frac{m!}{(2m)!} z^{2m}, \qquad p_m(z) = 1 + q_m(z). \tag{29}$$

These are even polynomials with real coefficients and $p_m(0) = 1$. Since $\binom{2m}{m} \ge 1$, one has $(m!)^2 \le (2m)!$. Thus, on $|z| \le R$,

$$|q_m(z)| \le \frac{R^{2m}}{m!} \longrightarrow 0.$$

The convergence follows, for example, because these are terms of the convergent exponential series $\sum_m R^{2m}/m!$. Consequently $p_m \to 1$ uniformly on every compact subset of $\mathbb{C}$. The same estimate also gives the common, but overly large, bound $|p_m(z)| \le 1 + e^{|z|^2}$.

Gaussian integration by parts with density $e^{-y^2/4}/\sqrt{4\pi}$ yields

$$\int y^{2m}\,d\nu_1(y) = 2(2m-1)\int y^{2m-2}\,d\nu_1(y) = \frac{(2m)!}{m!}.$$

The boundary terms vanish by polynomial Gaussian decay; iteration starts with mass one. On the imaginary axis the two signs cancel:

$$q_m(iy) = \frac{m!}{(2m)!}y^{2m} \geq 0.$$

Every polynomial is absolutely integrable there, but nevertheless

$$\mathcal{H}q_m = 1, \qquad \mathcal{H}p_m = 2, \qquad \mathcal{H}1 = 1, \qquad \int |p_m(iy) - 1|\,d\nu_1(y) = 1. \tag{30}$$

Hence $\mathcal{H}$ is not sequentially continuous for compact-open convergence, even on normalized even real polynomials. At the polynomial level the same value is obtained by the finite differential expression

$$(e^{-\partial_z^2}p_m)(0) := \sum_{j=0}^{m}\frac{(-1)^j}{j!}p_m^{(2j)}(0) = 2 :$$

only $j = 0, m$ contribute. This equality is a finite algebraic calculation, not a definition of that exponential on arbitrary entire functions.

There is no common $g \in L^1(\nu_1)$ dominating $|p_m(iy)|$ almost everywhere for every m. Otherwise countability permits a common exceptional-null-set removal and dominated convergence would force the integrals to tend to one, contrary to (30). A further elementary human consequence locates the missing mass: for every fixed R, compact convergence gives $\int_{|y|\leq R} q_m(iy)\,d\nu_1 \to 0$, whereas the total integral is one. Its complementary integral therefore tends to one. Pointwise convergence has lost uniform integrability at infinity.

For comparison, a sufficient condition is a common bound $|f_m(iy)| \leq Ce^{\tau y^2}$ with $\tau < 1/4$, together with compact convergence of f_m to f. The same bound holds for the limit, and its product with the Gaussian density is a constant multiple of $e^{-(1/4-\tau)y^2}$, which is integrable. Dominated convergence then does give $\mathcal{H}f_m \to \mathcal{H}f$. This is a labelled classical sufficient condition, not a necessary growth classification or an additional named formal theorem. The polynomials (29) have not been asserted real-rooted. They therefore do not refute any theorem with an additional split-polynomial or growth hypothesis. The example isolates the missing topology assumption; it changes none of the actual order-limit or zero conclusions proved in this paper.

10 Formal correspondence and editorial status

All principal results are supported by the frozen commit

`bab688bb6140cd6316c80680095a914ea0d346db.`

The namespace prefix used below is `OperatorDerivativeRewrite.Foundation`. Exact source links are in `theorem-map.json`, together with line numbers and SHA256 hashes at this commit.

Paper content	*Lean namespace and declaration*

Kernel, actual integral	`RiemannQuarticKernel.kernel`; `RiemannRealOrderFlow.flow_eq_cosine_integral`
Lemma 2.2	`RiemannRealOrderFlow.flow_differentiable`, `actual_flow_zero_pos`; `RiemannRealOrderComplexZeros.actual_subquadratic_growth`
First fractional range	`RiemannRealOrderFractionalCoefficient.actual_flow_scaled_limit`
Second fractional range	`RiemannRealOrderFourthAsymptotic.actual_flow_scaled_limit`
Cubic endpoint	`RiemannRealOrderCubicEndpoint.actual_flow_scaled_limit`
Lemma 4.1	`SubquadraticFiniteZeroFactorization.finite_zero_exponential_polynomial`, `exponential_polynomial_not_real_decay`
Theorem 1.1	`RiemannRealOrderQuarticLimitZeros.between_one_two_real_zeros_finite`, `between_one_two_nonreal_zeros_infinite`, `between_one_two_no_eventual_real_zeros`
Corollary 4.2	`RiemannRealOrderQuarticLimitZeros.between_one_two_no_split_polynomial_limit`
Order convergence	`RiemannRealOrderLimit.flow_order_tendsto_locally_uniformly`; `RiemannRealOrderRealUniform.flow_order_tendsto_uniformly_real`
Lemma 6.1, Proposition 6.2	`RiemannRealOrderZeroLimits.derivative_zero_of_nonreal_order_limit`, `eventually_real_zeros_on_compact`
Theorem 1.2	`RiemannRealOrderZeroLimits.nonreal_zero_norm_tendsto_atTop`; `RiemannRealOrderFractionalZeros.exists_approaching_escaping_nonreal_zeros`
(17)	`RiemannRealOrderFractionalCoefficient.leading_div_order_gap_limit`

Section 8 is mapped to the uniform endpoint, regular-part, tail and cutoff bounds, the actual transition-rate theorems and the rectangular relative obstruction. The explicit Bell-polynomial and reciprocal-profile estimates are human reconstructions of that verified mechanism; their constants are not claimed to be optimized. The new uniform Fourier and zero-radius conclusions rest on the full analytic proofs, not a numerical frequency scan.

Section 9 uses the actual Gaussian averages in `RiemannQuadraticGaussian` and the polynomial, integrability and nonconvergence statements in `GaussianHeatTopology`. Their exact declarations are in the theorem map. The compact-tail mass and sufficient Gaussian-growth criterion are fully proved classical human deductions. Thirty-six exact finite checks verify moment, sign and normalization transcriptions; they do not establish the analytic limits.

The elementary pointwise estimates in the kernel and canceled-remainder proofs are human reconstructions of the verified mechanism. They are not claimed as additional named Lean declarations with the same big-O notation. The actual derivatives, integrability, endpoint limits, integration-by-parts identities and final convergence statements have their own exact declarations. In particular, the fourth-order cancellation uses a separately verified actual fifth derivative; a formal exponential jet defined only through order four is not differentiated as if it were that fifth derivative. The density obstruction is separately recorded as `RiemannRealOrderBranchObstruction.three_halves_actual_density_obstruction`.

The baseline audit records a complete successful root build and 54,943 declarations audited with only `propext`, `Classical.choice`, and `Quot.sound`. The editorial pass does not rerun a redun-

dant full Lean build or change mathematical sources. Hash-linked provenance, compilation logs, and visual-review receipts accompany this draft.

Pólya's original 1927 discussion [2, Chapter I, no. 6, assertion II, pp. 14–15] already proves the smooth odd-endpoint obstruction: infinitely many zeros in a fixed horizontal strip force every odd endpoint derivative to vanish. For $e^{tu^3}K(u)$ the third derivative is $6tK(0) \neq 0$. Thus finiteness of its real zeros is a classical consequence; the coefficient in (14) is made explicit here. The nonintegral densities do not satisfy that statement's smoothness premise.

The contribution is the precise arithmetic statement and its readable proof, with exact endpoint and limit safeguards. General endpoint Fourier asymptotics, finite-zero rigidity, Hurwitz's theorem, and the role of simple zeros are established analysis. Priority for the arithmetic specialization remains unresolved. A publication decision should compare it directly with Pólya-type non-even-power transforms and subsequent real-zero-preserving transform criteria.

References

[1] G. Pólya, On the zeros of an integral function represented by Fourier's integral, *Messenger of Mathematics* **52** (1923), 185–188. Primary paper identified in the preliminary literature search; detailed historical theorem comparison remains pending.

[2] G. Pólya, *Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen*, Det Kgl. Danske Videnskabernes Selskab, Mathematisk-fysiske Meddelelser **7**, no. 17 (1927), 3–33. Primary text.

[3] D. A. Cardon, Fourier transforms having only real zeros, *Proc. Amer. Math. Soc.* **133** (2005), 1349–1356 (published electronically October 18, 2004). doi:10.1090/S0002-9939-04-07677-4.

[4] NIST Digital Library of Mathematical Functions, *Integrals of a real variable*, §2.3(i)–(ii). Official reference. Derivative-variation and endpoint-method context only.